

ME101: Engineering Mechanics, 2023-24

IIT Guwahati

Tutorial – 12 (Solution) on 22.04.2024

Q1: The uniform slender bar of mass ρ per unit length is freely pivoted about the y axis at the clevis (Fig.1), which rotates about the fixed vertical z -axis with a constant angular velocity ω . Determine the steady-state angle θ assumed by the bar. Length b is greater than length c .
[15 Marks]

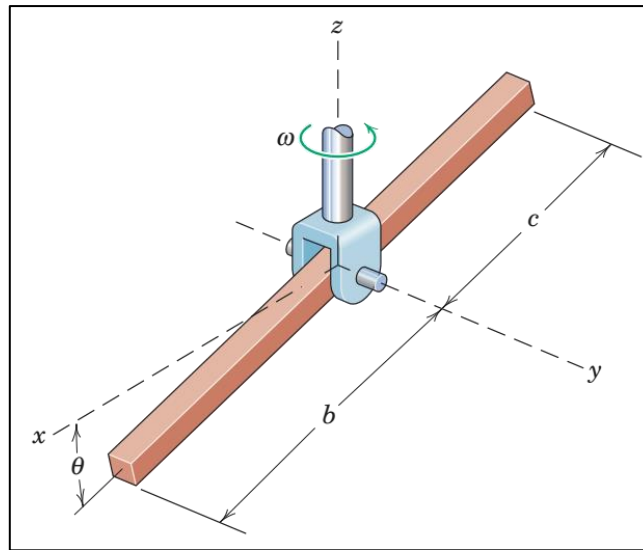


Figure 1

Solution:

$Oxyz$ is fixed reference frame and $Gx'y'z'$ is the rotating reference frame attached to the rod where G is the mass center of the rod (Fig. 2).

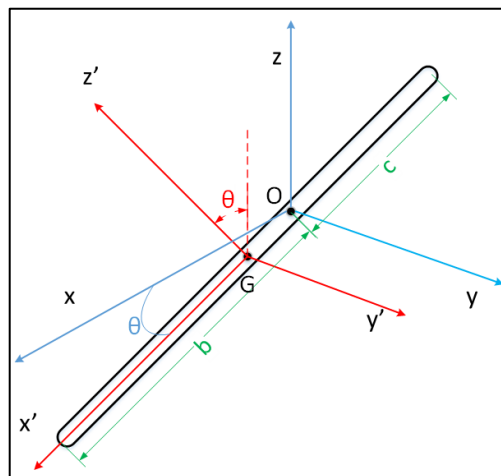


Figure 2

Let, $OG = d = \frac{b-c}{2}$ and length of the rod is $l = b + c$.

$$\text{Mass moment of inertia tensor in } Gx'y'z', \mathbf{I}_{Gx'y'z'} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & \frac{1}{12}ml^2 & 0 \\ 0 & 0 & \frac{1}{12}ml^2 \end{bmatrix}$$

Let, $\hat{\mathbf{i}}, \hat{\mathbf{j}}, \hat{\mathbf{k}}$ are unit vectors along co-ordinate axes in Oxyz and $\hat{\mathbf{i}}, \hat{\mathbf{j}}, \hat{\mathbf{k}}$ are unit vectors along co-ordinate axes in $Gx'y'z'$.

$$\text{Now, } \mathbf{r}_{G/O} = d(\cos\theta\hat{\mathbf{i}} - \sin\theta\hat{\mathbf{k}})$$

$$\mathbf{a}_G = \omega\hat{\mathbf{k}} \times \omega\hat{\mathbf{k}} \times \mathbf{r}_{G/O} = -\omega^2 d \cos\theta\hat{\mathbf{i}} \quad [2 \text{ Marks}]$$

$$\boldsymbol{\omega} = \omega\hat{\mathbf{k}} = \omega(\cos\theta\hat{\mathbf{k}} - \sin\theta\hat{\mathbf{i}})$$

$$(\mathbf{H}_G)_{Gx'y'z'} = \mathbf{I}_{Gx'y'z'} \boldsymbol{\omega} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & \frac{1}{12}ml^2 & 0 \\ 0 & 0 & \frac{1}{12}ml^2 \end{bmatrix} \begin{bmatrix} -\omega\sin\theta \\ 0 \\ \omega\cos\theta \end{bmatrix} = \frac{1}{12}ml^2\omega\cos\theta\hat{\mathbf{k}} \quad [3 \text{ Marks}]$$

$$\text{As } \boldsymbol{\omega} \text{ is constant, } \dot{\boldsymbol{\omega}} = 0. \text{ Then, } (\dot{\mathbf{H}}_G)_{Gx'y'z'} = 0 \quad [1 \text{ Marks}]$$

Now, the frame $Gx'y'z'$ is rotating with angular velocity $\boldsymbol{\omega}$.

$$\begin{aligned} \text{So, } (\dot{\mathbf{H}}_G)_{Oxyz} &= (\dot{\mathbf{H}}_G)_{Gx'y'z'} + \boldsymbol{\omega} \times (\mathbf{H}_G)_{Gx'y'z'} \\ &= \omega(\cos\theta\hat{\mathbf{k}} - \sin\theta\hat{\mathbf{i}}) \times \frac{1}{12}ml^2\omega\cos\theta\hat{\mathbf{k}} \\ &= \frac{1}{12}ml^2\omega^2\cos\theta\sin\theta\hat{\mathbf{j}} \quad [3 \text{ Marks}] \end{aligned}$$

$$\begin{aligned} \text{Now, } (\dot{\mathbf{H}}_O)_{Oxyz} &= (\dot{\mathbf{H}}_G)_{Oxyz} + \mathbf{r}_{G/O} \times m\mathbf{a}_G \\ &= \frac{1}{12}ml^2\omega^2\cos\theta\sin\theta\hat{\mathbf{j}} + d(\cos\theta\hat{\mathbf{i}} - \sin\theta\hat{\mathbf{k}}) \times m(-\omega^2 d \cos\theta)\hat{\mathbf{i}} \\ &= \left(\frac{1}{12}ml^2\omega^2\cos\theta\sin\theta + m\omega^2 d^2 \cos\theta\sin\theta \right) \hat{\mathbf{j}} \quad [3 \text{ Marks}] \end{aligned}$$

$$\begin{aligned} \text{Now, } \sum \mathbf{M}_O &= \mathbf{r}_{G/O} \times (-mg\hat{\mathbf{k}}) = d(\cos\theta\hat{\mathbf{i}} - \sin\theta\hat{\mathbf{k}}) \times (-mg\hat{\mathbf{k}}) \\ &= mgd\cos\theta\hat{\mathbf{j}} \quad [1 \text{ Marks}] \end{aligned}$$

$$\text{We know, } \sum \mathbf{M}_O = (\dot{\mathbf{H}}_O)_{Oxyz}$$

$$\text{Or, } mgd\cos\theta = m \left(\frac{l^2}{12} + d^2 \right) \omega^2 \cos\theta \sin\theta$$

Now, putting $l = b + c$ and $d = \frac{b-c}{2}$ in above expression we get,

$$\sin\theta = \frac{3g}{2\omega^2} \frac{b^2 - c^2}{b^3 + c^3}$$

$\therefore$ The steady state angle, $\theta = \sin^{-1}\left(\frac{3g}{2\omega^2} \frac{b^2 - c^2}{b^3 + c^3}\right)$ [2 Marks]

Q2. The position of the stylus tip A is controlled by the robot shown in Fig. 3. In the position shown the stylus moves at a rate of speed $u = 0.2$ m/s which reduces at a rate of 0.3 m/s² relative to the solenoid BC . At the same time, arm CD rotates at the constant rate $\omega_2 = 3$ rad/s with respect to component DEG . Knowing that the entire robot rotates about the X -axis at rate $\omega_1 = 5$ rad/s increases at the rate of $\alpha_1 = 2$ rad/s² determine (a) the velocity of A , (b) the acceleration of A . Take $L_1 = 750$ mm, $L_2 = 1000$ mm, $L_3 = 500$ mm, $L_4 = 300$ mm. [15 Marks]

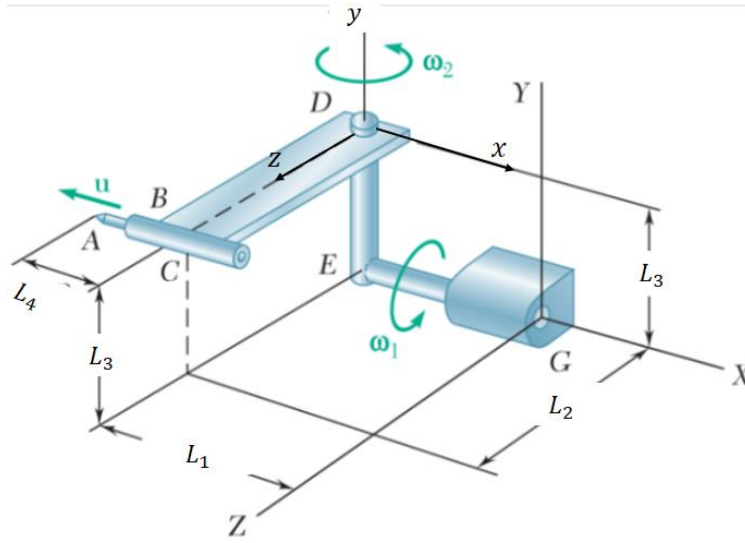


Figure 3

Solution:

Given, $L_1 = 750$ mm, $L_2 = 1000$ mm, $L_3 = 500$ mm, $L_4 = 300$ mm, $u = 0.2$ m/s, $\omega_1 = 5$ rad/s, $\alpha_1 = 2$ rad/s², $\omega_2 = 3$ rad/s.

Let, we have frame x - y - z which is parallel to the X - Y - Z , its origin is at D , it is attached to link CD .

Now, we know,

$$\vec{V}_A = \vec{V}_D + \vec{\Omega} \times \vec{r}_{A/D} + \vec{V}_{rel} \quad \dots (1)$$

$$\text{Here, } \vec{V}_D = \text{Absolute velocity of } D = \omega_1 \hat{i} \times 0.5 \hat{j} = 0.5 \omega_1 \hat{k} = 2.5 \hat{k}$$

$$\vec{\Omega} = \omega_1 \hat{i} + \omega_2 \hat{j} = 5 \hat{i} + 3 \hat{j}$$

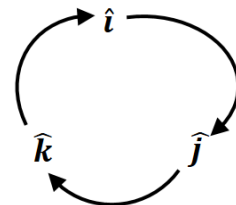
$$\vec{r}_{A/D} = \hat{k} - 0.3 \hat{i}$$

$$\vec{\Omega} \times \vec{r}_{A/D} = (5 \hat{i} + 3 \hat{j}) \times (\hat{k} - 0.3 \hat{i})$$

$$= -5 \hat{j} + 3 \hat{i} + 0.9 \hat{k}$$

$$\dots (B) \quad [2 \text{ Marks}]$$

$$\dots (A) \quad [1 \text{ Marks}]$$



$$\overrightarrow{V_{rel}} = \text{Relative velocity of } A \text{ with respect to } x\text{-}y\text{-}z \text{ frame} = -0.2\hat{i} \quad \dots \text{(C)}$$

Using (A), (B) & (C) we can write (1) as,

$$\overrightarrow{V_A} = 2.5\hat{k} - 5\hat{j} + 3\hat{i} + 0.9\hat{k} - 0.2\hat{i} = 2.8\hat{i} - 5\hat{j} + 3.4\hat{k} \text{ m/s} \quad [3 \text{ Marks}]$$

This is absolute velocity of point A.

Now, for acceleration we have,

$$\overrightarrow{a_A} = \overrightarrow{a_D} + \dot{\overrightarrow{\Omega}} \times \overrightarrow{r_{A/D}} + \overrightarrow{\Omega} \times (\overrightarrow{\Omega} \times \overrightarrow{r_{A/D}}) + 2\overrightarrow{\Omega} \times \overrightarrow{V_{rel}} + \overrightarrow{a_{rel}} \quad \dots \text{(2)}$$

$\overrightarrow{a_D}$ = Absolute acceleration of point D

$$= \omega_1 \hat{i} \times \omega_1 \hat{i} \times 0.5\hat{j} + 0.5\alpha_1 \hat{k}$$

$$= -0.5\omega_1^2 \hat{j} + 0.5\alpha_1 \hat{k}$$

$$= -12.5\hat{j} + \hat{k} \quad \dots \text{(D)} \quad [1 \text{ Marks}]$$

$$\overrightarrow{\Omega} = \omega_1 \hat{i} + \omega_2 \hat{j}$$

$$\dot{\overrightarrow{\Omega}} = \dot{\omega}_1 \hat{i} + \dot{\omega}_2 \hat{j} + \omega_2 \dot{\hat{j}}$$

$$\dot{\hat{j}} = \omega_1 \hat{i} \times \hat{j} = \omega_1 \hat{k}$$

$$\dot{\omega}_1 = \alpha_1 = 2 \text{ rad/s}, \quad \dot{\omega}_2 = 0,$$

$$\therefore \dot{\overrightarrow{\Omega}} = 2\hat{i} + \omega_1 \omega_2 \hat{k}$$

$$\dot{\overrightarrow{\Omega}} \times \overrightarrow{r_{A/D}} = (2\hat{i} + \omega_1 \omega_2 \hat{k}) \times (\hat{k} - 0.3\hat{i})$$

$$= -2\hat{j} - 0.3\omega_1 \omega_2 \hat{j}$$

$$= -2\hat{j} - 4.5\hat{j}$$

$$= -6.5\hat{j} \quad \dots \text{(E)} \quad [2 \text{ Marks}]$$

$$\overrightarrow{\Omega} \times (\overrightarrow{\Omega} \times \overrightarrow{r_{A/D}}) = (5\hat{i} + 3\hat{j}) \times (-5\hat{j} + 3\hat{i} + 0.9\hat{k}) \quad [\text{From B}]$$

$$= -25\hat{k} - 4.5\hat{j} - 9\hat{k} + 2.7\hat{i}$$

$$= 2.7\hat{i} - 4.5\hat{j} - 34\hat{k} \quad \dots \text{(F)} \quad [2 \text{ Marks}]$$

$$2\overrightarrow{\Omega} \times \overrightarrow{V_{rel}} = 2(5\hat{i} + 3\hat{j}) \times (-0.2\hat{i}) = 1.2\hat{k} \quad \dots \text{(G)} \quad [2 \text{ Marks}]$$

$$\overrightarrow{a_{rel}} = \text{Relative acceleration of point A w. r. t frame } x\text{-}y\text{-}z = 0.3\hat{i} \quad \dots \text{(H)} \quad [1 \text{ Marks}]$$

Using (D), (E), (F), (G), (H), we get from (2),

$$\overrightarrow{a_A} = -12.5\hat{j} + \hat{k} - 6.5\hat{j} + 2.7\hat{i} - 4.5\hat{j} - 34\hat{k} + 1.2\hat{k} + 0.3\hat{i}$$

$$\Rightarrow \overrightarrow{a_A} = 3\hat{i} - 23.5\hat{j} - 31.8\hat{k} \text{ m/s}^2 \quad [1 \text{ Marks}]$$

This is absolute acceleration of point A.

Q3. The centre O of the spacecraft is moving through space with a constant velocity of $8\hat{k}$ m/s as shown in Fig. 4. During the period of motion prior to stabilization, the spacecraft has a constant rotational rate $\Omega = 5$ rad/sec about its z -axis. The x - y - z axes are attached to the body of the craft, and the solar panels rotate about the y -axis at the constant rate $\dot{\theta} = 2$ rad/sec with respect to the spacecraft. If ω is the absolute angular velocity of the solar panels, determine $\dot{\omega}$. Also find the velocity and acceleration of point A when $\theta = 30^\circ$. Take $L = 1$ m, $t = 0.2$ m. [10 Marks]

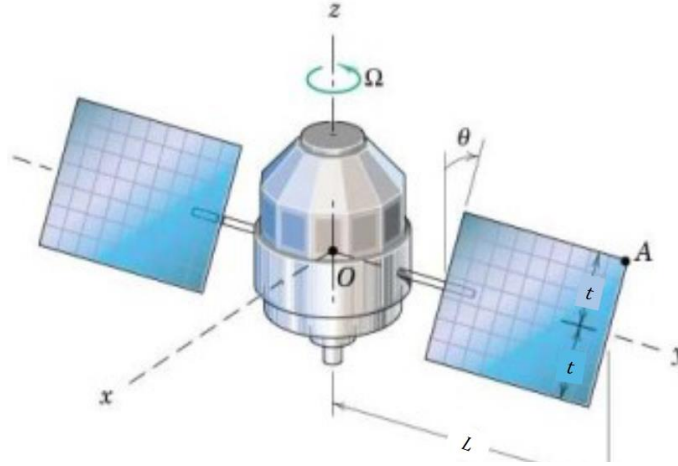


Figure 4

Solution:

Given, $L = 1$ m, $t = 0.2$ m, $\theta = 30^\circ$, $\Omega = 5$ rad/s, $\dot{\theta} = 2$ rad/s.

Here, axis x - y - z is rotational.

Absolute angular velocity of solar panel, $\vec{\omega} = \Omega\hat{k} - \dot{\theta}\hat{j} = 5\hat{k} - 2\hat{j}$ rad/s

Now, $\dot{\vec{\omega}} = \dot{\Omega}\hat{k} + \Omega\dot{\hat{k}} - \ddot{\theta}\hat{j} - \dot{\theta}\dot{\hat{j}}$

Here, Ω and $\dot{\theta}$ are constant $\therefore \dot{\Omega} = \ddot{\theta} = 0$

$\dot{\hat{k}} = \Omega\hat{k} \times \hat{k} = 0$; $\dot{\hat{j}} = \Omega\hat{k} \times \hat{j} = -\Omega\hat{i}$

$\therefore \dot{\vec{\omega}} = \dot{\theta}\Omega\hat{i} = 10\hat{i}$ (Ans)

Now, $\vec{V}_A = \vec{V}_O + \vec{\Omega} \times \vec{r}_{A/O} + \vec{V}_{rel}$

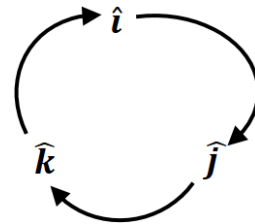
Here, $\vec{V}_O = 8\hat{k}$ m/s

And $\vec{r}_{A/O} = -0.2 \sin 30^\circ \hat{i} + 0.2 \cos 30^\circ \hat{k} + \hat{j} = -0.1\hat{i} + \hat{j} + 0.1732\hat{k}$

Also, $\vec{V}_{rel}$ = Relative velocity of A w.r.t. x - y - z frame

$\vec{V}_{rel} = -2\hat{j} \times (-0.2 \sin 30^\circ \hat{i} + 0.2 \cos 30^\circ \hat{k}) = -0.2\hat{k} - 0.3464\hat{i}$

$\therefore \vec{V}_A = 8\hat{k} + 5\hat{k} \times (-0.1\hat{i} + \hat{j} + 0.1732\hat{k}) - 0.2\hat{k} - 0.3464\hat{i}$



$$= 8\hat{k} - 0.5\hat{j} - 5\hat{i} - 0.2\hat{k} - 0.3464\hat{i} = -5.35\hat{i} - 0.5\hat{j} + 7.8\hat{k} \text{ m/s} \quad (\text{Ans})$$

[4 Marks]

$$\text{Now, } \vec{a}_A = \vec{a}_O + \dot{\vec{\Omega}} \times \vec{r}_{A/O} + \vec{\Omega} \times (\vec{\Omega} \times \vec{r}_{A/O}) + 2\vec{\Omega} \times \vec{v}_{rel} + \vec{a}_{rel}$$

Here, $\vec{a}_O = 0$ as, $\vec{v}_O$ is constant and $\dot{\vec{\Omega}} = 0$ as, Ω is constant, $\dot{\vec{k}} = 0$

$$\vec{\Omega} \times \vec{\Omega} \times \vec{r}_{A/O} = 5\hat{k} \times 5\hat{k} \times (-0.1\hat{i} + \hat{j} + 0.1732\hat{k}) = 5\hat{k} \times (-0.5\hat{j} - 5\hat{i}) = 2.5\hat{i} - 25\hat{j}$$

$$\vec{a}_{rel} = -\dot{\theta}^2(-0.2 \sin 30^\circ \hat{i} + 0.2 \cos 30^\circ \hat{k}) = 0.4\hat{i} - 0.6928\hat{k}$$

$$2\vec{\Omega} \times \vec{v}_{rel} = 2 \times 5\hat{k} \times (-0.2\hat{k} - 0.3464\hat{i}) = -3.46\hat{j}$$

$$\therefore \vec{a}_A = 2.5\hat{i} - 25\hat{j} - 3.46\hat{j} + 0.4\hat{i} - 0.6928\hat{k}$$

$$= 2.9\hat{i} - 28.46\hat{j} - 0.69\hat{k} \text{ m/s}^2 \quad (\text{Ans}) \quad [6 \text{ Marks}]$$

Q4. A solid cone of height h (300 mm) with a circular base of radius r (100 mm) is supported by a ball-and-socket joint at A as shown in Fig. 5. Knowing that the cone is observed to precess about the vertical axis AC at the constant rate of 40 rpm in the sense indicated and that its axis of symmetry AB forms an angle $\beta = 40^\circ$ with AC, determine the rate at which the cone spins about the axis AB. [10 Marks]

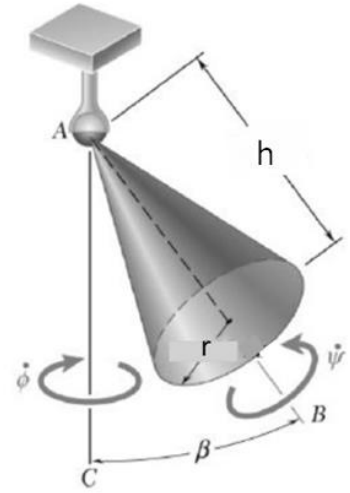


Figure 5

Solution:

Using principal axis xyz with origin at A

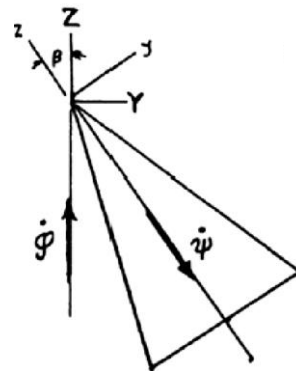
For a solid cone,

$$c = \frac{3h}{4}$$

$$I = I_{zz} = \frac{3}{10} \cdot mr^2$$

$$I' = I_{xx} = I_{yy} = \frac{3}{5} \cdot m \left(h^2 + \frac{r^2}{4} \right)$$

$$I' - I = \frac{3}{5} \cdot m \left(h^2 - \frac{r^2}{4} \right)$$



Angular velocity

Spin: $\dot{\psi}$ about negative z axis

Precession: $\dot{\phi}$ about positive z axis

$$\omega = \dot{\phi}\hat{k} - \dot{\psi}\hat{k}$$

$$\omega = \dot{\phi}(\cos \beta \hat{k} + \sin \beta \hat{j}) - \dot{\psi} \hat{k}$$

$$\omega_x = 0, \omega_y = \dot{\phi} \sin \beta, \omega_z = \dot{\phi} \cos \beta - \dot{\psi}$$

Angular momentum about fixed point A

$$H_A = I' \omega_x \hat{i} + I' \omega_y \hat{j} + I \omega_z \hat{k}$$

$$H_A = I' \dot{\phi} \sin \beta \hat{j} + I(\dot{\phi} \cos \beta - \dot{\psi}) \hat{k}$$

Frame Axyz be rotating with angular velocity Ω

$$\Omega = \dot{\phi} \hat{K} = \dot{\phi} \cos \beta \hat{j} + \dot{\phi} \sin \beta \hat{k} \quad [2 \text{ Marks}]$$

Rate of change of angular momentum H_A

$$\begin{aligned} \dot{H}_A &= \Omega \times H_A = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & \dot{\phi} \sin \beta & \dot{\phi} \cos \beta \\ 0 & I' \dot{\phi} \sin \beta & I(\dot{\phi} \cos \beta - \dot{\psi}) \end{vmatrix} \\ &= - \left[(I' - I) \dot{\phi}^2 \cos \beta \sin \beta + I \dot{\phi} \dot{\psi} \sin \beta \right] \hat{i} \quad [4 \text{ Marks}] \end{aligned}$$

Momentum about A

$$M_A = -mgc \sin \beta \hat{i} \quad [1 \text{ Marks}]$$

$M_A = \dot{H}_A$ leads to

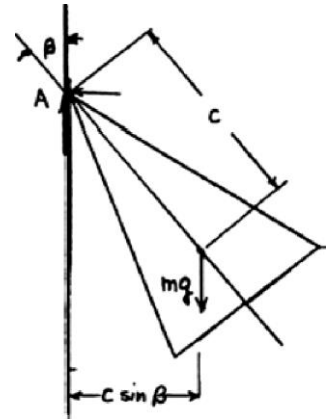
$$\begin{aligned} gc &= \frac{(I' - I)}{m} \dot{\phi}^2 \cos \beta + \frac{I}{m} \dot{\phi} \dot{\psi} \sin \beta \\ &= \frac{3}{5} \left(h^2 - \frac{r^2}{4} \right) \dot{\phi}^2 \cos \beta + \frac{3}{10} r^2 \dot{\phi} \dot{\psi} \end{aligned}$$

$$20gc = (12h^2 - 3r^2) \dot{\phi}^2 \cos \beta + 6r^2 \dot{\phi} \dot{\psi}$$

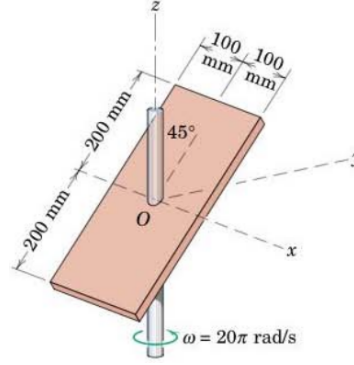
$$\dot{\psi} = \frac{20gc - (12h^2 - 3r^2) \dot{\phi}^2 \cos \beta}{6r^2 \dot{\phi}}$$

Putting data,

$$\dot{\psi} = \frac{(20)(9.81)(0.225) - [(12)(0.3)^2 - (3)(0.1)^2](4.188)^2 \cos 40^\circ}{6(0.1)^2(4.188)} = 119.493 \text{ rad/s} \quad [3 \text{ Marks}]$$



Q5. The plate, shown in Figure 5, has a mass of 3 kg and is welded to the fixed vertical shaft, which rotates at the constant speed of 20π rad/s. Compute the moment $\mathbf{M}$ applied to the shaft by the plate due to dynamic imbalance.



Solution 5: We can see, its a case of parallel plane motion, for which moment equations become,

$$\begin{aligned}\sum M_x &= -I_{xz}\dot{\omega}_z + I_{yz}\omega_z^2 \\ \sum M_y &= -I_{yz}\dot{\omega}_z - I_{xz}\omega_z^2 \\ \sum M_z &= I_{zz}\dot{\omega}_z\end{aligned}$$

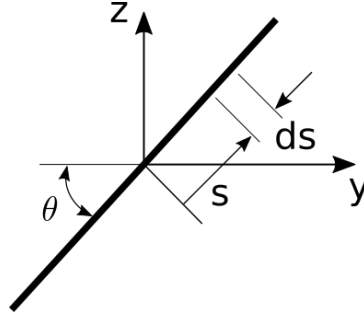


Figure 1: Side view of the plate

Note, $\omega_x = \omega_y = 0$, $\omega_z \neq 0$, $\dot{\omega}_z = 0$ and $I_{xz} = 0$. Mass per unit length of the plate, $m_0 = \frac{3}{0.4} = 7.5$ kg/m. Referring the side view of the plate,

$$\begin{aligned}I_{yz} &= \int_{-0.2}^{0.2} s \cos \theta s \sin \theta m_0 ds & [3 \text{ marks}] \\ &= \frac{1}{2} m_0 \sin 2\theta \int_{-0.2}^{0.2} s^2 ds \\ &= 0.02 \text{ kg-m}^2 & [2 \text{ marks}]\end{aligned}$$

$$\sum M_x = I_{yz}\omega_z^2 = 78.96 \text{ N-m, (it is applied on the plate)} \quad [3 \text{ marks}]$$

$$\sum M_y = \sum M_z = 0 \quad [1 \text{ mark}]$$

Thus, the moment applied to the shaft is $-78.96\hat{i}$ N-m. [1 mark]